Venkata Lakshmi Parimala Pasupuleti

Data Engineer · Python · SQL · AWS · ETL Pipelines · Distributed Data · ML Pipelines · Data Modeling

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PROFESSIONAL SUMMARY

Data Engineer with 5+ years at Entrata designing and building **production data pipelines, ETL systems, and distributed data architectures** across regulated, high-volume environments. Built and maintained pipelines processing 1.6M+ HUD/Tax Credit records, optimized 200+ SQL queries (30% performance gain across 1M+ records), and architected async distributed processing for multi-tenant data at scale. At Intuit, building **ML pipelines** (scikit-learn, XGBoost) from feature engineering through scoring APIs and integrating data quality validation into AI/LLM production workflows. Strong Python, SQL, AWS (S3, Lambda, RDS, SQS), Docker, Kubernetes, and Terraform IaC. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

Intuit

Jun 2026 – Present

Software Engineer — ML Pipelines · Data Engineering · AI Data Quality

- Build **ML pipelines** (scikit-learn, XGBoost) end-to-end: feature engineering, data validation, model training, scoring APIs, and integration with downstream financial data services on AWS (Lambda, S3, RDS, SQS).
- Design and implement **data quality validation frameworks** for LLM and ML pipeline outputs; build golden-dataset regression suites ensuring data accuracy and consistency before every production release.
- Implement data security and PII-safe handling patterns; deploy pipelines via Docker and GitHub Actions CI/CD; instrument with Prometheus and OpenTelemetry for **data observability**.
- Partner with cross-functional Agile teams to translate business data requirements into scalable, maintainable pipeline architectures.

George Mason University

Aug 2024 – Aug 2025

Research Associate — Data Engineering · Statistical Modeling · Analytics

- Built and cleaned multi-source datasets (WHO global health + UK Road Safety, 190+ countries, 104K+ records); engineered features and built regression models achieving **$R^2 = 0.977$** with full data lineage.
- Applied Isolation Forest anomaly detection across 3 research pipelines; designed self-service BI dashboards (Tableau, ggplot2) improving research decision-making **40%** across two cohorts.

Entrata — Residential & Affordable Housing SaaS

Oct 2019 – Jul 2024

Senior Data Engineer — ETL Pipelines · Distributed Data · Data Modeling

- Designed and maintained **ETL pipelines processing 1.6M+ HUD/Tax Credit records** -- federal API ingestion, data transformation into ~180,000 domain records, validation, reconciliation, and rollback at each release cycle. Zero data integrity failures over 5 years.
- Architected **RabbitMQ-based distributed async data processing** for rent and eligibility updates across thousands of tenant records -- idempotency guarantees, retry logic, and dead-letter handling for consistent multi-tenant data at scale.
- Built and optimized **200+ SQL queries** via execution plan analysis, achieving 30% performance improvement across 1M+ property records; designed complex joins, conditional logic, stored procedures, and multi-schema data models across PostgreSQL and Oracle.
- Led **AWS S3 data migration** with Libsodium PHI encryption covering 100% of resident data; deployed data services via Docker microservices on AWS (S3, Lambda, RDS) with Terraform IaC.
- Built Python/Bash automation eliminating **80% of manual data pipeline workload**; enforced data quality standards, naming conventions, and audit lineage across multi-source datasets; 100% PII compliance, zero security incidents.
- Collaborated with Agile engineering and product teams on end-to-end data solution design; reviewed 300+ PRs; mentored junior engineers on data pipeline patterns and database optimization.

PROJECTS

Leasel: Compliance Data Pipeline with End-to-End Lineage (1,240 Records)

Python · SQL · LangGraph · SQLite · FastAPI · Audit Logging — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built end-to-end data pipeline with full audit lineage, data quality validation at each stage, and traceable outputs — **93.6% precision, 87.9% F1** across 1,240 test cases.

CrashClarity: Large-Scale Data Engineering & ML Pipeline (104K Records)

Python · SQL · Scikit-learn · Random Forest · Tableau — Live Dashboard

- Designed full data engineering pipeline: ingestion, cleaning, feature engineering, distributed ML model training (85%+ accuracy), and self-service Tableau dashboard deployed to end users.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — May 2026

GPA: 3.87 · Fairfax, Virginia · Distributed systems, ML pipelines, data governance, statistical modeling, anomaly detection

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University — 2017

TECHNICAL SKILLS

Languages & Data Tools: Python (Pandas, NumPy, scikit-learn, XGBoost, PySpark), SQL (advanced joins, stored procedures, execution plan analysis), Scala (familiar), Java, R

Data Pipeline & Engineering: ETL Pipeline Design, Distributed Data Processing, Data Modeling (relational + non-relational), RabbitMQ Async Processing, REST API Integrations, Feature Engineering, Data Quality & Observability, Audit Lineage

Cloud & Infrastructure: AWS (S3, Lambda, RDS, SQS, EMR-adjacent), Terraform (IaC), Docker, Kubernetes, GitHub Actions CI/CD, Prometheus, OpenTelemetry

Databases & Warehousing: PostgreSQL, Oracle, SQLite, MongoDB (familiar), DynamoDB (familiar), Snowflake (familiar), Redshift (familiar)

ML & Analytics: ML Pipelines, Model Evaluation & Scoring, Anomaly Detection (Isolation Forest), Tableau, Power BI, Agile/Scrum, Cross-Functional Collaboration